Abdullah Al Minhaz
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Education

Bachelor of Science in Electrical and Electronic Engineering (EEE) *January 2023 – Present*
(3rd Year, 2nd Semester running)
Jamalpur Science & Technology University, Jamalpur, Bangladesh
CGPA: 3.50 (till 2nd Year)
Currently holding provisional 1st position in class (based on completed semesters)
Previously awarded Merit Scholarship (2nd position) in 1st Year Examination.

Higher Secondary Certificate (HSC) — Science, 2021 | GPA: 5.00
Secondary School Certificate (SSC) — Science, 2019 | GPA: 5.00

Research Interests

- Biomedical Instrumentation and Medical Device Development
- Embedded Systems and Real-Time Data Acquisition
- Intelligent Sensing, Signal Processing, and Applied Artificial Intelligence
- Control Systems and Robotics
- Power Systems and Renewable Energy Systems

Personal Statement

Electrical and Electronic Engineering student with a strong research interest in biomedical instrument development, physiological signal processing, and intelligent diagnostic systems. Experienced in control systems, robotics, and embedded system–based data acquisition and analysis. Seeking higher studies to work on computational modeling, medical device design, and AI-assisted biomedical systems for next-generation healthcare technologies.

University Projects (Hardware, Embedded & Systems)

Autonomous Self-Driving Vehicle Platform *September 2023 - Present*

- Developed a Raspberry Pi 5–based embedded vision and control system implementing lane detection and maneuver planning using discrete-time control in Python.
- Implemented a YOLOv8n-based visual event detection module integrated with motor control to enforce safety-critical 5-second full-stop behavior.

RFID & Google Sheets–Based Smart Attendance System *August 2025*

- Developed an ESP8266-based RFID data acquisition and cloud logging system with real-time synchronization to Google Sheets.
- Implemented NTP-based timestamping and local LCD feedback for reliable embedded data recording.

PID Line Following Robot *May 2024 – June 2024*

- Designed and implemented a **PID-based closed-loop motor control system** for accurate real-time line tracking.
- Tuned and validated the controller to improve tracking accuracy and dynamic response.

Design, Setup and Analysis of Off-Grid Solar System *May 2024*

- Designed and implemented a 0.5 kW off-grid photovoltaic power system integrating PV array, MPPT charge controller, battery storage, and inverter.
- Evaluated system performance, battery state-of-charge behavior, and inverter efficiency under real operating conditions.

Simple Line Following Robot *June 2023*

- Built an autonomous robot using IR sensors and microcontroller-based motor control; implemented a basic line-following algorithm.

Software & Data Analysis Projects

SQL Case Study: Google Play Store Dataset *April 2025*

- Performed data cleaning and exploratory analysis on Google Play Store app data using Python and MySQL.
- Implemented advanced SQL techniques including CTEs, triggers, and stored procedures to analyze ratings, revenue, and category-wise performance.

SQL Case Study: Data Science Jobs & Salary Trends *April 2025*

- Conducted a comprehensive data analysis of the global data science job market using MySQL.
- Used window functions and analytical queries to study salary trends, remote work adoption, company size distribution, and experience-level pay differences.

Technical Skills

Programming & Data:

- Python, C, MATLAB, SQL, MySQL

Embedded Systems & Hardware:

- Raspberry Pi, ESP8266 (NodeMCU), Arduino, RFID RC522, LCD (I2C), Motor Drivers

AI, Data Science & Computer Vision:

- Machine Learning, Deep Learning, OpenCV, YOLO, TensorFlow/Keras, Exploratory Data Analysis

Power, Control & Simulation:

- MATLAB/Simulink, PID Control, Power System Analysis, Solar Energy Systems

Tools & Platforms:

- Git, Linux, Arduino IDE, Google Colab, LaTeX, Microsoft Word/Excel/PowerPoint, Google Apps Script

Other Skills:

- Data Cleaning, Stored Procedures, Window Functions, Technical Presentation, Team Leadership, Communication

Certifications & Training

- Introduction to Artificial Intelligence — Simplilearn (2023)

- Ethical Hacking 101 — Simplilearn (2023)
- Design, Setup and Analysis of Off-Grid Solar System — RERC, JSTU

Self-Directed Learning & Online Training

- **OpenCV–Python: Zero to One (Professional Computer Vision)** — Aryan Verma (Infoaryan)
  Completed an in-depth professional training series on computer vision using OpenCV and Python, covering image processing, filtering, feature extraction, edge detection, segmentation, morphological operations, and project-based implementations with strong theoretical foundations.
- **100 Days of Machine Learning** — CampusX
  Completed a comprehensive, structured curriculum (134+ lessons) covering data preprocessing, exploratory data analysis, feature engineering, supervised and unsupervised learning, model evaluation, and applied machine learning workflows using Python.
- **100 Days of Deep Learning** — CampusX
  Completed an end-to-end deep learning curriculum (80+ lessons) covering artificial neural networks (ANN), convolutional neural networks (CNN), recurrent neural networks (RNN), optimization techniques, regularization, and practical implementation using TensorFlow/Keras.

Code Repositories & Technical Portfolio

- **GitHub:** https://github.com/abdullahalminhaz
  Contains implementations of machine learning, deep learning, computer vision, embedded systems, SQL data analysis, and robotics projects, including academic projects and self-study experiments.
- **Kaggle:** https://www.kaggle.com/alminhaz
  Includes data analysis notebooks, machine learning experiments, feature engineering pipelines, and dataset exploration projects.

Academic Activities & Competitions

- Participant, Robotics Exhibition, National Conference, JSTU
- Participant, 1st Inter-Department Mathematics Olympiad 2023, JSTU
- Participant, Project Showcase, Department of EEE, JSTU

Awards & Scholarships

- Merit Scholarship Award (2022) — Secured 2nd position in 1st Year B.Sc. Engineering Examination, JSTU
- 2nd Place, Farm Mechanization Idea Competition (2025) — JSTU & FMDP, BARI

Leadership & Volunteering

- **Research Team Coordinator (Founding Committee)** — JSTU Research Society *(Sep 2025 – Present)*
- **Secretary General** — Chattogram Divisional Student Welfare Council, JSTU *(Dec 2025 – Present)*
- **Class Representative (CR)** — Jamalpur Science & Technology University *(Mar 2024 – Mar 2025)*
- **Secretary of Information & Technology** — EEE Association, JSTU *(Nov 2022 – Nov 2023)*
- **Deputy Secretary of Publicity** — Chattogram Divisional Student Welfare Council *(Oct 2022 – Oct 2023)*
- **Secretary of Education** — Hridoy Memorial Blood and Social Foundation *(Jun 2020 – Jul 2021)*
- **Volunteer** — GST Admission Test 2024–2025
- **Volunteer, Volunteer Monitoring Unit** — 9th University Day, JSTU *(Nov 2025)*